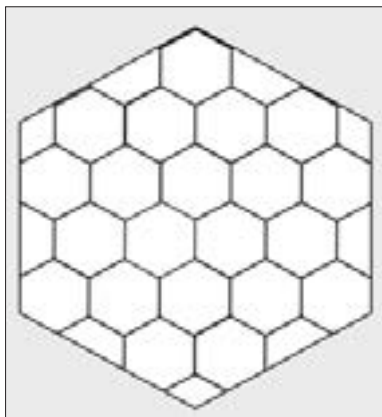


gles!), the hexagon emerges the clear victor. This way, the problem of interference between two cells was avoided using only four different frequency bands. A CDMA network, however, doesn't need to bother with this; more on this later in this chapter.

Now that we have our cellular network working, it's time to bust a misconception: speaking about cells being serviced by a single base-station is a little inaccurate. This conjures up the image of a hexagon with an antenna at the centre, which is entirely erroneous. In reality, a base station is located at the corner of a hexagon, servicing three cells; rather, it services one-third of each of these three cells. In the cellular network then, each hexagonal cell has three base-stations at its corners.

Networks Within Networks

We have seen how breaking up a network into cells increases the number of callers that can be handled by the network. In urban areas, cells can be further divided into pico-cells or hot-spots, taking the capacity of the network another step ahead. This means that in city centres, where cellular traffic usually puts a strain on the network, more calls can be handled. In addition, because your mobile phone is even closer to its base station, you can get a strong signal even in unventilated corners of a building.



Lots of pico-cells inside a larger cell

You And Your Network

Each operator has its own *Mobile Telephone Switching Office (MTSO)*, which handles all the phone connections on a network and con-